EPSR-test-ex94 nsmoop 0, refpotfac 0.8, rbroad 0.05

Program(s): epsr | chained passes: 3 | verification: ALL IDENTICAL (all 3 passes)

What this case tests

Exercises three tuning options of the `epsr` engine of EPSR [1,2]: switching off the subtraction of a smooth background from the data (`nsmoop 0`), scaling the reference potential (`refpotfac`) and the exponential decay `rbroad` of the Poisson basis functions of the empirical potential [3].

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `runepsr17` (`nsmoop`: `ftback` not called)
- `epsr_standalone_rev3.f`: `inita` (reading `refpotfac`, `rbroad`), `generate_rp` (reference LJ + Coulomb scaled by `refpotfac`)
- `epsr_standalone_rev3.f`: `generate_ep`, `calc_pwrexpr_grad` (`exp(-rbroad*r)` and gradient factor `n/r-1/sigma-rbroad`)

Rust (port, rust/epsr-core/src/):

- `epsr.rs`: `Epsr::setup`, `Epsr::iterate` (`nsmoop`); `structure.rs`: `ftback`
- `structure.rs`: `inita`; `potential.rs`: `generate_rp`
- `potential.rs`: `generate_ep`, `calc_pwrexpr_grad`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. For this case the Lennard-Jones and Coulomb reference potentials are multiplied by 0.8, every Poisson basis function of the EP decays additionally as exp(-0.05·r), and the data are fitted without background subtraction.

Inputs

Made from EPSR-test with `scripts/make_case.py`: `.EPSR.inp` `nsmoop 0`; `.pcof` `refpotfac 0.80000` and `rbroad 0.05000000`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 3 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 3 passes.
The values are written back to the `.pcof` and `.EPSR.inp` and read again in the next pass, including the list-directed single-precision form of `rbroad` (`5.00000007E-02`) and `refpotfac` written with format `(f8.5)`, which are reproduced exactly. With `nsmoop 0` the difference between data and simulation enters `ptgnsq` without the `ftback` correction, and the scaled reference potential enters every Monte Carlo energy and move decision.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.